

# Kun-Hsiang Lin

[Email](#) | [Website](#) | [Linkedin](#) | [GitHub](#)

## Professional Summary

Ph.D. candidate in Computer Science at National Taiwan University with 7+ years of experience spanning academic research and industrial AI development. Specialized in biometric AI, face anti-spoofing, and vision-language large models, with demonstrated impact in top-tier research venues and real-world deployed systems.

## Experience

**Senior AI Engineer / AI Team Lead**, Authme Oct. 2019 – Feb. 2025

- Designed and deployed production-grade CV and biometric AI systems across mobile, web, and edge platforms, delivering proprietary algorithms for face recognition, anti-spoofing, OCR, and fraud detection.
- Built cross-platform C++ SDKs (Android, iOS, WebAssembly) enabling low-latency, stable inference, reducing feature integration time from 3 months to 3 weeks.
- Established end-to-end MLOps pipelines for data curation, training, deployment, and online monitoring, cutting model development cycles from 3 months to 2 weeks.
- Led AI engineers as team lead, aligning research, product, and regulatory requirements; achieved ISO/IEC 30107-3 compliance and #1 NIST FRVT 1:1 ranking in Taiwan.

**Research Assistant**, DMID Lab, IIS, Academia Sinica Sep. 2017 – Sep. 2019

- Developed deep spatiotemporal models for typhoon rainfall nowcasting on real-world climate datasets.
- Supported peer review for top-tier conferences (NeurIPS, ICML, KDD).

**Research Assistant**, Hydraulic and Ocean Engineering, NCKU Jul. 2017 – Aug. 2017

- Implemented a drought early-warning system in C for local government use.

## Education

**National Taiwan University**, Ph.D. in Computer Science and Information Engineering Sep. 2024 – Present

- **Research fields:** Computer Vision, Domain Generalization, Anomaly Detection, VLMs, MLLMs
- **GPA:** 4.15 / 4.3 (Advisor: [Wen-Huang Cheng](#))

**National Taiwan University**, M.S.E. in Computer Science and Information Engineering Sep. 2023 – Jun. 2024

- **GPA:** 4.12 / 4.3, Rank#1 (Advisor: [Wen-Huang Cheng](#))

**National Chen Kung University**, M.S.E. in Hydraulic and Ocean Engineering Sep. 2014 – Jun. 2016

- **Thesis:** [Comparison of SVM and RF for Hourly Typhoon Rainfall Forecasting](#)
- **GPA:** 3.81 / 4.0, Rank#1 (Advisor: [Pao-Shan Yu](#))

**National Chen Kung University**, B.E. in Hydraulic and Ocean Engineering Sep. 2010 – Jun. 2014

## Awards & Highlights

### International Competitions & Challenges

- 2024 – NIST FRTE 1:1 (USA): 44/371 globally, 1st in Taiwan
- 2023 – CVPR 4th Face Anti-Spoofing Challenge: 16/66 globally
- 2021 – ICCV Masked Face Recognition Challenge: 5/160 globally, 1st in Taiwan

### Certifications & Recognitions

- 2023 – FIME ISO Verification: Achieved ISO/IEC 30107-3 compliance; 1st in Taiwan
- 2023 – Taiwan AI Award, AI Taiwan Future Conference

### Industry PoC & Hackathons

- 2020 – Taiwan RegTech eKYC Challenge: Group Champion
- 2019–2021 – Achieved Top 1 ranking in eKYC PoCs by Cathay Financial, Standard Chartered, and Line Bank

### Academic Awards & Scholarship

- 2024 – Ministry of Education Ph.D. Scholarship: National-level funding for outstanding doctoral students
- 2016 – Chi-Hsin Agricultural Development Foundation Scholarship

## Technologies

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### Machine Learning & AI

- Frameworks & Libraries: PyTorch, Lightning, TensorFlow, ONNX, ONNXRuntime
- Expertise: Computer Vision, Biometric AI (Face), Anomaly Detection, VLMs, MLLMs, Digital Image Processing, Model Optimization, Distributed Training, Edge ML, MLOps

### Software Engineering

- Languages & Tools: Python, C/C++ , Bash Scripting, GitFlow, Docker
- Practices: DevOps, Test-Driven Development, Algorithm Design & Implementation

### Research & Leadership

- Experience: AI Algorithm Innovation, Research Prototyping, Team Management

## Publications

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### **InstructFLIP: Exploring Unified Vision-Language Model for Face Anti-spoofing**

**Lin, K. H.**, Tseng, Y. W., Huang, K. Y., Wu, J. C., Cheng, W. H.

ACM International Conference on Multimedia, Oct. 2025 — Oral, Top-tier in Computer Science and Multimedia

### **Predictor Selection Method for the Construction of SVM-based Typhoon Rainfall Forecasting Models using a Non-dominated Sorting Genetic Algorithm**

Yang, T. C., Yu, P. S., **Lin, K. H.**, Kuo, C. M., Tseng, H. W.

Meteorological applications, Oct. 2018 — SCI-indexed Journal

### **An Acceptable Framework to Predict the Flood Stage Under Climate Change Scenarios - A Case Study in Taipei Basin**

Wu, P. Y., **Lin, K. H.**

Asia Oceania Geosciences Society Annual Meeting, Jul. 2017 — Poster, Top-tier in Hydraulic Engineering

### **A Comparison of Hourly Typhoon Rainfall Forecasting Models Based on Support Vector Machines and Random Forests with Different Predictor Sets**

**Lin, K. H.**, Tseng, H. W., Kuo, C. M., Yang, T. C., Yu, P. S.

EGU General Assembly, Apr. 2016 — Oral, Top-tier in Hydraulic Engineering

### **The Application of Support Vector Machine and Random Forest on Precipitation Forecasting**

**Lin, K. H.**, Tseng, H. W., Kuo, C. M., Yang, T. C., Yu, P. S.

Conference on Computer Applications in Civil and Hydraulic Engineering, Sep. 2015 — Best Student Paper Award

### **A Comparison of Random Forests and Support Vector Machine in River Stage Forecasting**

**Lin, K. H.**, Tseng, H. W., Kuo, C. M., Yang, T. C., Yu, P. S.

Asia Oceania Geosciences Society Annual Meeting, Jul. 2015 — Oral, Top-tier in Hydraulic Engineering

## Projects

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### **Capybara: Python Toolkit for Computer Vision Task**

- Developed an image processing and deep learning toolkit with modules for vision tasks, structured data handling, ONNX inference, and test coverage.

### **Chameleon: Python Toolkit for Developing AI Models**

- Designed a modular DL framework with reusable components and training utilities for rapid prototyping.

### **DocClassifier: Document Image Classification System**

- Built a metric learning-based classification system enhanced with ImageNet-1K and CLIP embeddings; achieved 90%+ accuracy with high inference speed and scalability.

### **FaceDetection: Efficient Face Detection Training Module**

- Built on SCRFD with custom enhancements, the face-detection-XL model achieved SOTA on WIDERFace with mAPs of 0.965 (Easy), 0.951 (Medium), and 0.845 (Hard), offering strong efficiency.

### **TWLLM-Tutor: Revolutionizing Taiwanese Secondary Education with Large Language Model**

- Developed an AI-based education system using a customized GSAT dataset and LLM tuning methods to support learning in underserved regions.